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Who says statistics classes are boring? How a virtual escape room enhances statistics learning

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ABSTRACT

Escape rooms (ERs) have gained traction in higher education over the past decade, but studies on ERs for statistics education are scant. In this study, a virtual ER was developed, piloted, and implemented in an undergraduate introductory statistics class, and its impact on learning was evaluated over three iterations. The escapED framework informed the design of the ER. A quasi-experimental pre-test–post-test design was used to evaluate learning gains. A post-activity questionnaire was used to collect student perceptions about the game. Significant learning gains were observed, with large effect sizes across all cohorts. Similarly, analyses of the qualitative data revealed that the game enhanced learning in the cognitive, affective, behavioural, and interpersonal domains, and was well-received by students. The ER was effective in improving learning outcomes. ERs can be a powerful approach in fostering peer learning and shifting negative attitudes about difficult subjects. Educators keen in exploring ERs in their classrooms are highly encouraged to consider pedagogically-informed game designs and leverage technology in creating engaging and effective learning experiences.

1. Introduction

In a world that is increasingly data-driven, statistical literacy has become an essential skill across diverse industries. Statistical literacy, defined as “the ability to understand and critically evaluate statistical results” encountered in everyday life (Wallman, 1993), is now a fundamental competency that educational institutions must impart to students, with skills such as interpreting data and appraising statistical claims taking precedence. Yet, despite its importance, statistics is often perceived negatively by students as dry, difficult (Griffith et al., 2012), irrelevant, and anxiety-inducing (Onwuegbuzie & Wilson, 2003), hindering students’ engagement with statistics learning. Engagement in statistics is multifaceted and complex - affective, behavioural, and cognitive engagement are involved in the learning process (Whitney et al., 2019) and facilitated by self-efficacy (Linnenbrink & Pintrich, 2003). Bandura (1997) describes self-efficacy as "beliefs in one’s capabilities to organise and execute the courses of action required to produce given attainments”. These beliefs play a crucial role in motivating students to engage in effective studying and learning behaviours – in contrast, maladaptive beliefs about one’s ability contribute to low self-efficacy, which in turn leads to low motivation, and a lack of interest in the subject (Boyle et al., 2014; Bromage et al., 2022; Onwuegbuzie & Wilson, 2003). Students with low self-efficacy often perceive statistical tasks as more challenging than they actually are and are less likely to ask for help (Ryan et al., 2001). This

Abbreviations: ER, escape room; GAISE, Guidelines for Assessment and Instruction in Statistics Education; GBL, game-based learning; STEM, science, technology, engineering, and mathematics.

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reluctance can result in poorer academic performance (Ryan et al., 2001) which may further entrench negative attitudes toward learning statistics.

1.1. Pedagogical underpinnings

Garfield and Ben-Zvi (2007) developed eight evidence-based principles about learning statistics rooted in constructivism (Vygotsky, 1978), active learning (Bonwell & Eison, 1991), and experiential learning (Dewey, 1938).

1. Students learn by constructing knowledge.
2. Students learn by active involvement in learning activities.
3. Students learn to do well only what they practice doing.
4. It is easy to underestimate the difficulty students have in understanding basic concepts of probability and statistics.
5. It is easy to overestimate how well students understand basic concepts.
6. Learning is enhanced by having students become aware of and confront their errors in reasoning.
7. Technological tools should be used to help students visualise and explore data, not just to follow algorithms to pre-determined ends.
8. Students learn better if they receive consistent and helpful feedback on their performance.

In line with these principles, the Guidelines for Assessment and Instruction in Statistics Education (GAISE) was developed for a practical application of these principles in statistics courses (Carver et al., 2016).

1. Teach statistical thinking.
    i. Teach statistics as an investigative process of problem-solving and decision making.
    ii. Give students experience with multivariable thinking.
2. Focus on conceptual understanding.
3. Integrate real data with a context and purpose.
4. Foster active learning.
5. Use technology to explore concepts and analyse data.
6. Use assessments to improve and evaluate student learning.

As statistics is not generally known to be a subject that appeals ubiquitously to students, statistics educators face the age-old challenge of fostering enthusiasm and motivation to engage learners in their classes. In their seminal work, Bonwell and Eison (1991) defined approaches that foster active learning as "instructional activities involving students in doing things and thinking about what they are doing". Such activities require higher-order thinking, allow students to challenge misconceptions, interrogate intra-personal attitudes and values, and often involve collaboration in groups (Freeman et al., 2014), enabling students to learn more effectively compared to traditional didactic approaches.

1.2. Game-based learning

Active learning can take on diverse forms – of note, GAISE recommends using games as a way to incorporate active learning and technology in statistics education. While gamification focuses on using incentives to motivate learners without modification to the learning activity itself, in game-based learning (GBL), a learning activity is redesigned as a game ("serious game") to enhance its effectiveness for learning (Plass et al., 2020). Although it can be argued that all games present learning opportunities (e.g., gaining technical knowledge or developing motor skills), a key difference in game-based learning is the explicit function of the "serious game" in helping students (1) gain new knowledge and skills on significant topics, (2) reinforce existing knowledge and abilities, (3) foster learning and innovation skills, and/or (4) prepare for future learning (Plass et al., 2020). Collaborative serious games also facilitate the development of interpersonal skills (López-Pernas, 2024; Taraldsen et al., 2022). Reviews on GBL in statistics education (e.g., Boyle et al., 2014; Lekka et al., 2017) have also suggested that GBL can be an effective approach for enhancing statistics learning, underscoring a potential for its further application in university education.

1.3. Escape rooms (ERs) in education

Since its debut in 2013, escape rooms (ERs) have become increasingly popular as a tool in education, with games designed specifically for learning (Nicholson, 2015; Taraldsen et al., 2022). ERs were traditionally implemented in physical settings as a recreational activity, where players work collaboratively to "escape" one or more locked rooms by finding clues to solve puzzles and/or perform tasks under time constraints (Nicholson, 2015). Educational ERs, however, adopt a socio-constructivist approach (Vygotsky, 1978) – students construct knowledge during the gameplay experience and collaborate with their team members alongside guidance from the teacher to tackle in-game tasks. Educational ERs have been shown to enhance enjoyment, satisfaction, motivation and fun among students (López-Pernas, 2024). Variations of the escape room concept have been reported, such as escape boxes (Tidbury & Bridge, 2023; Veldkamp et al., 2020a) and decoder boxes (Ross & Hall, 2021). However, in view of scalability considerations for large classes, the development of virtual educational ERs has been further propelled by the shift to online learning during the COVID-19 pandemic due to logistical, space, cost, and time constraints in implementing physical ERs (Makri et al., 2021; Quek et al., 2024).

1.3.1. Challenges with using ERs in education and mitigation measures

Despite being a creative approach to teaching and learning, the use of ERs is not without challenges. Implementing ERs can be resource-intensive in terms of logistics, space, time and cost (Fotaris & Mastoras, 2019), for example, implementation in large classes may be logistically difficult. Designing ERs can also be an intellectual challenge, specifically in balancing the mechanical and pedagogical aspects of the game (Taraldsen et al., 2022). In addition, there are inconsistencies in methodology and rigour in ER research on its effectiveness in promoting learning. Recommendations to address these challenges include the need for well-designed ERs that are constructively aligned with learning objectives, using more systematic approaches to assess the learning effectiveness of ERs, conducting the studies over several iterations, and incorporating post-game debriefing (López-Pernas, 2024; Quek et al., 2024; Veldkamp et al., 2020b).

1.4. Rationale and research question

The use of ERs in education has been recently reviewed (Burbage & Pace, 2024; Hintze et al., 2023; Lathwesen & Belova, 2021; López-Pernas, 2024; Makri et al., 2021; Quek et al., 2024; Taraldsen et al., 2022; Veldkamp et al., 2020b). Most educational ERs were implemented in higher education settings, with some in secondary and primary education, and a majority implemented in healthcare disciplines (e.g., medicine, nursing, pharmacy) or STEM (science, technology, engineering, mathematics) subjects. Despite the rise in popularity of ERs in higher education, there are, however, few published studies on ERs for learning statistics. Students who participated in an ER-inspired puzzle activity showed increased content knowledge and self-efficacy post-game and had higher final exam marks than non-participants (McIntyre, 2023). Besides this, Tidbury and Bridge (2023) reported positive student perceptions and perceived learning gains toward a physical “escape box” game for medical statistics.

Given the scalability issues with physical ER implementation in our educational context, we developed a virtual ER for an undergraduate introductory statistics class and explored its impact on student learning. This research study therefore aims to address the following question: How does a virtual ER impact student learning in introductory statistics?

2. Material and methods

2.1. Learning context

Ethical approval was obtained for this research study (NUS-IRB-2022-266). The ER was implemented for a single flipped session in an undergraduate introductory research methods course. The course is offered twice a year as a core course in the Minor and Second Major in Public Health programmes in a Singapore university. The course spans over 13 weeks (one semester), and aims to introduce students (in chronological order) to the research cycle, ethics and project management, and quantitative and qualitative approaches to public health research. Classes are 3-h long in a lecture-tutorial format and are held on campus once a week. Students are assigned to project groups in Week 3 to work on a group project culminating in a presentation and report due in Weeks 11–12 and 13 respectively.

Specifically, the quantitative methods section is conducted from Weeks 4–7, covering introductory statistics (e.g., Week 4: Types of quantitative data/variables, measures of central tendency, binomial, t- and normal distributions, graphical presentations of data, the central limit theorem, confidence intervals, sampling; Weeks 5 and 6: Hypothesis testing and inferential statistics [bivariate analysis]; Week 7: Practical session using statistical software). The summative assessment for this section is a quantitative assignment (released in Week 7), which requires students to use the appropriate statistical tests to address research questions related to a realistic public health dataset.

2.2. Design of the ER

The design of the ER activity was informed by Clarke et al.’s (2017) escapED framework, which has been widely adopted for ER development and evaluation (Burbage & Pace, 2024). This framework was developed based on the works by Arnab and Clarke (2017) and Nicholson (2015) to provide structural guidance in creating educational ERs and related games in higher education. The six key components of this framework are described in the following sub-sections (2.2.1 to 2.2.6).

2.2.1. Participants

Students taking the course (“participants”) are non-medical undergraduates from diverse disciplines (e.g., Biomedical Engineering, Psychology) and years of study (i.e., Years 1–5) with a range of statistical experience (naive to experienced). Table 1 shows the participant enrolment by faculty between the three cohorts. The majority of students were from the faculty of Science, followed by Engineering, and Arts and Social Sciences. There was no significant difference in faculty distribution between cohorts.

As the course sees large class sizes of more than 50 students, a virtual ER was developed due to scalability and resource constraints. A Web-based platform, Genially (https://genial.ly/), was used for the development of the ER due to the ease of use and interactive features suitable for a point-and-click educational game. Given the allotted time for face-to-face classes, iterative testing (section 2.2.6) indicated that the ER was best played cooperatively in teams of three to five to allow most groups to complete the game within 1 h 15 min.

Table 1
Participant enrolment by gender and faculty.

Gender	Cohort		
	Jan 2023	Aug 2023	Jan 2024
Male	19 (29%)	21 (34%)	13 (24%)
Female	47 (71%)	41 (66%)	42 (76%)
Faculty	**Cohort**		
	Jan 2023	**Aug 2023**	**Jan 2024**
Science	37 (56%)	31 (50%)	31 (56%)
Engineering	16 (24%)	14 (23%)	13 (24%)
Arts and Social Sciences	5 (8%)	6 (10%)	5 (9%)
Business	3 (5%)	5 (8%)	4 (7%)
Medicine (Nursing)	4 (6%)	6 (10%)	2 (4%)
Computing	1 (2%)	0	0
Total number of participants	66	62	55

2.2.2. Objectives

The learning objectives of the ER were aligned with those of the Week 4 lesson, which entailed foundational concepts that subsequent lectures would build upon up to the quantitative assignment in Week 11.

1. Distinguish between nominal, ordinal, ratio, and interval variables.
2. Distinguish between independent and dependent variables.
3. Present basic descriptive statistics.
4. Interpret a box plot.
5. Describe examples of events that follow a binomial distribution.
6. Determine probabilities using the t-distribution and normal distribution.
7. Understand the central limit theorem.
8. Understand how confidence intervals are constructed.
9. Describe basic sampling methods.

As students have been assigned to their project groups the preceding week (i.e., Week 3), the ER would also serve to facilitate team bonding for the subsequent group project and enhance the learning experience in the overall course.

2.2.3. Theme

The ER was designed with a whodunnit/murder mystery theme, in which players take on the role of detectives tasked to investigate a crime scene, gather evidence, and deduce the murderer's identity, motive, and means. An in-game screenshot of a case file is provided in Fig. 1. The ER narrative was introduced to students within the game itself, and sectioned into four chapters. Access to each chapter and most evidence is locked with a password. Players perform tasks and solve puzzles to obtain the passwords required to progress in the game until its completion.

2.2.4. Tasks and puzzles

The in-game tasks (Fig. 2) varied in difficulty level and were mapped with the learning objectives of Week 4's lesson (section 2.2.2, Table 2) to scaffold the knowledge and skills required to understand the subsequent weeks' topics on hypothesis testing and inferential statistics. Progressively, learning-related tasks ranged from quizzes on statistical concepts to more complex authentic tasks such as exploring, analysing, and visualising data from a realistic dataset (with an accompanying data dictionary) embedded in the game as part of the narrative. Among these tasks were two abstract puzzles in which players had to apply their understanding of quantitative data to solve. Many of these tasks were integrated into the theme and storyline of the game, for example, in unlocking the victim's laptop, and solving a statistics puzzle to match a fingerprint sample with those in a fingerprint database. Other tasks included searching for hidden clues in the virtual environment. Instructions and in-game hints for all tasks and puzzles were provided within the ER.

2.2.5. Equipment

The first author conceptualised and created the ER in early 2022. The ER was developed using an online Web platform (Genially) that is free to use. No physical props were required given the virtual nature of the game. Web resources were used to create some of the clues (e.g., fake receipts, a puzzle website, AI voice generator). To collect data on gameplay duration, two Google Forms were purposefully embedded in the ER as the first virtual lock (i.e., start time) and the penultimate virtual lock ("The Verdict"; i.e., end time). The last virtual lock was used to conceal the solution to the crime mystery ("The Truth"). Students successfully "escaped the tutorial" if they unlocked "The Truth". Google Drive was required for players to access the dataset and other files due to limited features in Genially for embedding these documents at the time of writing.

The computer laboratory was used as the game venue, considering the availability of desktop computers (until Dec 2023) that students could use, and its suitability for accommodating group seating arrangements for collaboration. PowerPoint slides were used

Table 2
Alignment of in-game tasks with learning objectives.

Learning objectives	In-game task(s)
Distinguish between nominal, ordinal, ratio, and interval variables.	1, 4
Distinguish between independent and dependent variables.	1, 3
Present basic descriptive statistics.	10, 12
Interpret a box plot.	10, 12
Describe examples of events that follow a binomial distribution.	1
Determine probabilities using the t-distribution and normal distribution.	1, 9
Understand the central limit theorem.	1, 7
Understand how confidence intervals are constructed.	1
Describe basic sampling methods.	6

for the game briefing and debriefing.

The total time taken to conceptualise and create the first prototype approximated 50 h, excluding beta-testing and subsequent optimisation. The second author provided inputs in the latter.

2.2.6. Evaluation

Prior to implementation, the ER was trialled iteratively, first among non-students, and then with students who have taken the course, to determine the optimum player size for a 1 h 15 min gameplay and to ensure that the game was pitched appropriately. The ER was piloted in the August 2022 cohort. Prior to each ER session, both authors checked through the game and rectified problems (if any) to ensure a smooth play-through. A post-activity questionnaire (section 2.4) was used to obtain student feedback regarding gameplay experience and whether the learning objectives were met through the ER activity. This feedback, along with the gamemasters' observations, were used to improve the gameplay for the subsequent cohorts. Pre- and post-tests were used to measure learning gains.

2.3. GBL implementation

Video lectures were created for Week 4's topics due to the flipped classroom design for this week. At the end of Week 3, students were tasked to view the video lectures before coming for the tutorial. They were, however, not made aware of the nature of the Week 4 face-to-face tutorial except to "come prepared" for the session[1].

At the start of the tutorial, the instructor (the first author) informed the class that the session's goal was to "escape the tutorial", and conducted a pre-game briefing (20 min) on the game instructions, game theme, time limit (1 h 15 min), availability of in-game hints, and other personal items required for the session (e.g., earphones, note-taking equipment), before releasing the URL and passcode to students to access the virtual ER. Students were seated in their project groups of four to five members and were allowed to play the ER in groups of three if there were absent team members. Each student played the game on their own device. Latecomers were given the option to join their own groups or form a new group of three to five. The gamemasters provided technical support on an ad-hoc basis, and students had the option to request for verbal hints from the gamemasters if they were still stuck after referring to the in-game hints. When verbal hints were requested, the gamemasters did not directly provide the solutions to students but instead guided them in independent problem-solving to maintain the integrity of the game objectives. Groups were told to stop all activity when the time was up. Students were allowed to help their group mates but not peers from other groups. They were, however, not allowed to leave the room except for toilet breaks, as a debrief was to be conducted when the 1 h 15 min was up. During the 30-min debrief, the instructor went through the solutions to the more challenging questions in the ER. During the briefing and after the debriefing, all students were reminded not to share any information about the ER with their peers outside of the course.

2.4. Study design

A quasi-experimental pre-test–post-test design was used to evaluate the impact of the ER on student learning. The ER was delivered as part of normal instruction to all students enrolled in the course from August 2022. In Week 1, a pre-ER quiz (pre-test) was administered on Canvas, the school's learning management system. Students who have "escaped" were tasked to complete the post-ER quiz (post-test) independently in class to assess learning gains, except the fourth cohort, to whom the post-ER quiz was administered one week after the ER (at the start of Week 5's lesson). The pre- and post-ER quizzes comprised nine multiple-choice questions and one multiple-select question according to the intended learning objectives of Week 4's content.

The post-activity questionnaire adapted from Gordillo et al. (2020) was administered after the ER session to gather demographic information, pre-course statistics experience, pre-ER gaming experience, and student perceptions towards the game. The quantitative section of the questionnaire comprised items on perceived learning, attitudes, gameplay experience, and collaborative learning. These items were rated on a Likert scale from 1 (Strongly Disagree) to 5 (Strongly Agree). The qualitative section included questions on key takeaways from the game, the most enjoyable and frustrating aspects of the game, and suggestions for game improvement.

[1] For the last cohort (i.e., Jan 2024), "coming prepared" included bringing their personal laptops since all desktop computers in the laboratory had been removed at the end of Dec 2023.

The data for the time taken by students to complete the game (defined as the time between unlocking the first and penultimate virtual locks), and the proportion of students who successfully "escaped" (defined as the students who had unlocked "The Truth"), were extracted from the Google Forms data.

As major refinements were made in the ER after the pilot run, only data collected from the subsequent three cohorts (Jan 2023, Aug 2023, Jan 2024) was used for this research study.

2.5. Data analysis

Descriptive statistics was used to summarise the quantitative data from the post-activity questionnaire and gameplay duration across cohorts. The Fisher's exact test was used to determine if there were any demographic differences between cohorts. The Shapiro-Wilk test was used to assess normality. Differences between pre-test and post-test scores within each cohort were analysed using the paired *t*-test and the Wilcoxon signed rank test. *P*-values <0.05 were considered statistically significant. Cohen's *d* was used as a measure of effect size. R Statistical Software (v4.0.4; R Core Team, 2023) was used for all statistical analyses.

The open-ended responses in the post-activity questionnaire were thematically analysed (Braun & Clarke, 2006) using a deductive-inductive approach. Given the multifaceted nature of learning, the lens of affective, behavioural, and cognitive (ABC) learning (Rogaten et al., 2019) was applied in the deductive analysis. The ABC model categorises learning gains as such: affective learning gains involve positive shifts in attitudes, satisfaction and well-being; behavioural gains entail acquisition of skills such as technical and soft skills, and cognitive gains refer to increase in cognitive-related abilities such as understanding, knowledge, critical and analytic thinking. The inductive approach was used to identify themes that did not fall within the ABC framework. Both authors conducted line-by-line coding and analysis independently using Taguette (desktop version) (Rampin & Rampin, 2021). The codebook

Table 3
Respondent demographics.

Faculty	Cohort			Total
	Jan 2023	Aug 2023	Jan 2024	
Science	29	14	26	69
Engineering	12	8	7	27
Arts and Social Sciences	5	1	4	10
Medicine (Nursing)	3	5	2	10
Business	2	2	2	6
Computing	1	0	0	1
Major	**Cohort**			**Total**
	Jan 2023	**Aug 2023**	**Jan 2024**	
Life Sciences	20	9	19	48
Biomedical Engineering	8	7	4	19
Nursing	3	5	2	10
Pharmaceutical Science	2	2	2	6
Data Science and Analytics	3	2	1	6
Pharmacy	2	1	1	4
Sociology	3	0	1	4
Chemical Engineering	2	0	1	3
Engineering	1	1	1	3
Social Work	0	1	2	3
Accountancy	0	1	1	2
Psychology	0	1	1	2
Statistics	0	0	2	2
Business	0	1	0	1
Business Administration	0	0	1	1
Chemistry	0	0	1	1
Civil Engineering	0	0	1	1
Communications and New Media	1	0	0	1
Environmental Biology	1	0	0	1
Food Science and Technology	1	0	0	1
Human Capital Management	1	0	0	1
Information Systems	1	0	0	1
Leadership and Human Capital Management	1	0	0	1
Materials Science Engineering	1	0	0	1
Year of Study	**Cohort**			**Total**
	Jan 2023	**Aug 2023**	**Jan 2024**	
1st year	12	11	15	38
2nd year	28	12	18	58
3rd year	9	4	2	15
4th or 5th year	3	3	6	12

was iteratively developed, and the final version was applied before the themes, sub-themes, and sub-sub-themes related to the main research question were identified. All analyses were cross-checked, discussed, and refined collectively between both authors. Any discrepancy was resolved by consensus. Illustrative quotes were extracted from the data for each sub-sub-theme.

3. Results

3.1. Respondent demographics

The demographics of respondents to the post-activity questionnaire are presented in Table 3. 123 students (response rate 67.2%) completed at least 40% of the questionnaire – this cut-off point was determined as it captured responses with 100% completion in the demographics and quantitative sections, forming the data that was analysed.

Similar to the cohort demographics, majority of respondents were from the Science faculty (56%), followed by Engineering (22%), and Arts and Social Sciences (8%). Most respondents read a primary major in Life Sciences (39%), followed by Biomedical Engineering (16%), and Nursing (8%), and were second-year (46%) or first-year (32%) students.

69% of respondents have taken an introductory statistics course before the ER session, and 46% have played an ER or a point-and-click game before the ER session. The majority of respondents (66%) rated the game difficulty as average, while 6.5% rated it easy and 27.6% difficult. Consistent across cohorts, the time taken to complete the game ranged from 37 min to 1 h 22 min, with more than 73% of respondents making a correct verdict. There was no significant difference in respondent demographics between the three cohorts, and given the small numbers in several categories, the differences in distribution of primary majors were not analysed.

3.2. Pre-test and post-test analyses

Table 4 presents the analyses of pre- and post-test quiz scores. Baseline pre-test scores were not significantly different between cohorts. Pre- and post-test scores, however, were significantly different ($p < 0.001$) in each cohort, indicating that all three cohorts demonstrated significant learning gains, with large effect sizes (Cohen's $d > 0.8$) (Cohen, 1988).

3.3. Post-activity questionnaire: quantitative section

Table 5 summarises the quantitative results of the questionnaire. Overall, the ER was well-received by students. Most students agreed that the game was aligned with Week 4's learning objectives (Q1) and was well-designed (Q10), while finding it to be an effective and more engaging approach for learning statistics compared to a traditional tutorial (Q2, 3, 7). Students also found the detective theme very engaging (Q11) and the game very enjoyable (Q4), yet remained neutral as to whether it was a stressful experience (Q5). Where group work was concerned, students enjoyed working with their team during the ER (Q6) and benefitted greatly from the teamwork (Q13), with consistent evidence of peer learning, support, and teamwork in the process (Q14, 16, 17). They felt that the assigned group size was adequate (Q18) and sufficient time was given for gameplay (Q12), as well as agreed that the in-game hints provided were useful (Q20) and the post-activity briefing was helpful (Q19). Given a choice, most students expressed a preference for playing the ER in a team (Q15). Interestingly, mastering the course materials posed a greater main difficulty to the gameplay for majority of students than overall game mechanics (Q8, 9).

3.4. Post-activity questionnaire: qualitative section

The qualitative data uncovered a nuanced interplay between the dynamics of ERs and learning outcomes that many studies have yet to delve deeper into. Analyses revealed a multitude of influences that shaped the learning outcomes of undergraduate public health students who have engaged with the ER to learn introductory statistics (Tables 6–8, Fig. 3).

Intrapersonal aspects emerged as the pillars upon which learning flourished – participants expressed profound shifts in cognitive and behavioural domains respectively, attributing areas such as the consolidation of learning ("*Reinforced the concepts learnt in class, eg boxplot, normal distribution etc.*" – *ID 109*) to the immersive nature of the ER, all while honing technical skills ("*By solving statistical problems, I learnt to apply what I have learnt from the lecture videos. For instance, finding interquartile range using google sheets.*" – *ID 60*) and soft skills ("*Teamwork is important, we need to share ideas and discuss to better understand certain concepts.*" – *ID 45*) in the process. Within the affective domain, students found the ER experience rewarding when they were "*able to solve the puzzles from using the concepts that* (they have) *learnt*" – *ID 27*. Notably, engagement in the ER was perceived to reduce the stress that students face when approaching new statistical content – an area often overlooked in traditional classroom delivery – with one student rightfully claiming, "... *the game was a less stressful way of learning concepts we need to grasp.*" – *ID 49*. An additional theme on attitudinal change was identified, in which the ER experience altered what students initially felt, believed, or assumed about statistics ("*Learning statistics can be in a more fun manner*" – *ID 6)*.

Interpersonal dynamics further enriched the GBL environment, fostering collaboration, camaraderie, and shared goals. As recounted by a participant, peer learning was key in fostering a deeper understanding of classroom content and mutual support among student learners, "*I liked that there was a storyline and that we could work together and help each other out in the game. If I did not know how to do a certain question, my friend would help me out and vice versa.*" – *ID 30*. As participants navigated in-game tasks, learning became a collective endeavour where individual abilities were recognised and exercised for a common end-goal while the learning process was found enjoyable, "*Different team mates have a stronger suit in different sections and we help each other and have fun together.*" – *ID 119*.

Table 4
Pre- and post-test analyses.

Cohort	Course enrolment	n[a]	Mean difference	SD	95% CI	*p*-value	Cohen's *d*
Jan 2023	66	54	2.26	1.86	(1.75, 2.77)	<0.001	1.22
Aug 2023	62	37	2.69	2.21	(1.95, 3.42)	<0.001	1.22
Jan 2024	55	37	1.91	1.38	(1.52, 2.30)	<0.001	1.66

[a] Missing data excluded.

From problem-solving to captivating narratives, the ER provided fertile social ground for exploration, discovery, and learning as a team.

Yet, amidst the potential of the ER, hurdles occasionally emerged, creating setbacks within the learning landscape. Specifically, issues that arose during gameplay, such as navigation within the game platform (e.g., "… *having to go back to different pages to look at the clues." – ID 16*), and technical issues of glitches/bugs (e.g., "*Some of the UI were missing ie. I could see the ending parts without clearing the probability questions that my groupmates faced." – ID 86*) hindered the learning experience for some. A few participants from the initial cohorts were also troubled by hardware issues from some of the laboratory computers, "*… the computer lab PC took forever to start and launch chrome." – ID 43*. Meanwhile, barriers to problem-solving, ranging from solution roadblocks (i.e., "*Not being able to come up with the code when we already had all the puzzle pieces in place." – ID 103*) to the effectiveness of hint systems (e.g., "*Sometimes the clues were hidden in plain sight." – ID 21*), posed challenges during the ER session. Apart from this, team dynamics, while fostering collaboration, also presented complexities as participants grappled with varied progress within groups, while external factors related to timing (e.g., "*The game takes place … when I am quite tired from the day of class." – ID 9*) and spatial limitations (e.g., "*… hearing answers from [other] groups spoil the fun*." – ID 52), occasionally disrupted the flow of learning, testing the resilience of participants' concentration and engagement in the ER activity.

However, despite these challenges for some, the GBL experience unfolded seamlessly without significant hurdles for others, *"… i dont think it was very frustrating, i enjoyed it!" – ID 142.* Overall, the efficacy of a carefully designed ER as an educational resource for learning introductory statistical concepts is best summed up by the following sentiment: *"The game is super super fun! It is a very enjoyable tutorial and it helped to recap a lot of the lecture content covered and apply it." – ID 144.*

4. Discussion

A virtual ER was developed, piloted, and implemented for a flipped classroom on introductory statistics, and its impact on learning was evaluated over three iterations. The quantitative and qualitative results consistently demonstrated that the ER enhanced learning multidimensionally across cohorts with diverse disciplinary backgrounds and statistical experience and was well-received by students overall. Analyses of pre- and post-tests showed significant learning gains by participating in the ER session, with large effect sizes observed in all three cohorts (Cohen's $d > 1.2$). This is consistent with the large effect size (Cohen's $d = 1.4$) reported in a recent meta-analysis of 33 educational ERs across healthcare, STEM, art, and education fields by López-Pernas (2024), who found that the effectiveness of ERs was consistent across settings, regardless of group size or technology used. Additionally, our findings aligned with McIntyre (2023)'s work, where increases in content knowledge were seen in both sessional groups after participating in an ER-inspired statistics review activity, although only one group demonstrated a statistically significant increase in content knowledge.[2]

Furthermore, the immersive nature of the ER provided a highly engaging experience for students. Features such as the game design, in-game interactivity, and captivating narrative allowed for greater cognitive and affective engagement and motivation among student learners as compared to traditional modes of delivery. Apart from this, the team-based gameplay instilled a sense of teamwork and agency in students through a shared goal of wanting to unlock "The Truth" and win the game. This invokes deep learning and may contribute to how ERs can engage and motivate student learners more effectively than traditional teaching approaches (Gee, 2008). Moreover, our qualitative findings suggest that the ER brought about other intangible benefits, such as creating a less intimidating environment for conceptual application where it is relatively "safe to fail" compared to being asked to respond to questions in class, and helping students "break the ice" with other group members to foster a positive group project experience, while allowing for an appreciation of statistics and its real-world relevance through situated learning in authentic contexts. However, some students continued to express sentiments that the main difficulty in the ER game lied in a mastery of course materials, exposing an underlying continued belief that mastering statistics remains difficult. Conversely, other findings suggest some evidence that negative attitudes toward statistics classes were altered through the ER activity, similar to a study by Glavaš and Stašcik (2017), where positive attitudes toward learning mathematics were observed to have increased after an ER game. Taken together, our findings suggest that the game design was effective in creating an immersive learning environment that not only enhanced engagement and enjoyment for students, but also fostered collaboration and peer learning, considering that statistics is a challenging subject for many. As attitudinal changes take time, it is hoped that continued ER activities would help alleviate perceived difficulties in mastering statistics over time.

Likewise, the themes in our study aligned with student perceptions toward an escape-box activity for medical statistics (Tidbury & Bridge, 2023). While students enjoyed the overall ER experience, however, they remained ambivalent regarding whether the gameplay

[2] Coupled with a small sample size, it was observed that the other group had a considerably high baseline content knowledge to begin with, which may account for the lack of statistical significance in content knowledge increase.

Table 5
Summary of perception ratings in the post-activity questionnaire.

Q	Item	Cohort	n	Mean (SD)	Median
1	The educational game aligns with the lecture learning objectives.	Jan 2023	41	4.23 (0.79)	4
		Aug 2023	51	4.35 (0.75)	4
		Jan 2024	29	4.35 (0.61)	4
2	I feel that I would learn more from this educational game than if the same learning objectives were to be covered in a traditional tutorial.	Jan 2023	40	3.73 (1.06)	4
		Aug 2023	41	3.94 (1.03)	4
		Jan 2024	51	4.05 (0.96)	4
3	The educational game was an effective way to reinforce the related topics covered in Week 4's lectures.	Jan 2023	29	4.29 (0.89)	4
		Aug 2023	40	4.32 (0.98)	5
		Jan 2024	41	4.37 (0.58)	4
4	I enjoyed the educational game.	Jan 2023	51	4.24 (0.88)	4
		Aug 2023	29	4.42 (0.92)	5
		Jan 2024	40	4.48 (0.55)	4.5
5	The educational game was a stressful experience.	Jan 2023	41	2.99 (1.12)	3
		Aug 2023	50	2.94 (1.26)	3
		Jan 2024	28	2.69 (1.00)	3
6	I enjoyed working with my team in the educational game.	Jan 2023	40	4.07 (0.87)	4
		Aug 2023	41	4.26 (0.86)	4
		Jan 2024	50	4.17 (0.66)	4
7	The educational game was more engaging than a traditional tutorial.	Jan 2023	28	4.32 (0.91)	4.5
		Aug 2023	40	4.45 (0.89)	5
		Jan 2024	41	4.45 (0.71)	5
8	The main difficulty of the educational game lied in mastering the course materials.	Jan 2023	50	3.62 (0.95)	4
		Aug 2023	28	3.55 (1.21)	4
		Jan 2024	40	3.48 (1.04)	4
9	The main difficulty of the game lied in the non-educational game mechanics.	Jan 2023	41	3.20 (1.19)	3
		Aug 2023	50	3.48 (1.15)	3
		Jan 2024	28	3.00 (1.23)	3
10	The educational game was well-designed.	Jan 2023	40	4.22 (0.86)	4
		Aug 2023	41	4.35 (0.84)	5
		Jan 2024	50	4.38 (0.58)	4
11	The detective-themed storyline was engaging.	Jan 2023	28	4.26 (0.83)	4
		Aug 2023	40	4.58 (0.56)	5
		Jan 2024	41	4.43 (0.63)	4.5
12	The time given for the educational game was adequate.	Jan 2023	50	4.35 (0.74)	4
		Aug 2023	28	4.52 (0.51)	5
		Jan 2024	40	4.40 (0.63)	4
13	I benefitted from the teamwork in my group.	Jan 2023	41	4.30 (0.86)	4
		Aug 2023	50	4.39 (0.84)	5
		Jan 2024	28	4.45 (0.67)	5
14	I learnt about statistics and/or quantitative research methods from my team member(s).	Jan 2023	40	4.11 (0.96)	4
		Aug 2023	41	4.13 (1.12)	4
		Jan 2024	50	4.14 (0.81)	4
15	I would rather prefer participating in the educational game alone than in a team.	Jan 2023	28	2.46 (1.19)	2
		Aug 2023	40	2.52 (1.26)	2
		Jan 2024	41	2.19 (1.02)	2
16	All the members of my team were equally involved in solving the different puzzles.	Jan 2023	50	4.04 (0.96)	4
		Aug 2023	28	4.29 (1.01)	5
		Jan 2024	40	3.88 (1.19)	4
17	My team mates helped each other out in the educational game.	Jan 2023	41	4.40 (0.77)	4.5
		Aug 2023	50	4.58 (0.56)	5
		Jan 2024	28	4.38 (0.66)	4
18	I would rather have been part of a larger team.	Jan 2023	40	2.28 (1.18)	2
		Aug 2023	41	2.13 (1.28)	2
		Jan 2024	50	2.26 (1.04)	2
19	The post-activity debrief was helpful.	Jan 2023	28	4.04 (0.79)	4
		Aug 2023	40	4.16 (0.78)	4
		Jan 2024	41	4.12 (0.74)	4
20	The hints provided within the game platform were useful to progress in the game.	Jan 2023	50	4.49 (1.01)	5
		Aug 2023	28	4.71 (0.97)	5
		Jan 2024	40	4.29 (0.86)	4

was a stressful experience. Possible sources of stress could include the time-imposed pressure for game completion and spending prolonged periods of time stuck on a puzzle, as evidenced in some qualitative responses in this study. The in-game hints were also found to be very useful, although there were some sentiments that these hints, along with the puzzle instructions, could have been more explicit. Such comments were mostly directed at one of the two abstract puzzles, which were incorporated in the game to moderate the difficulty level. Clearer in-game hints for the puzzle in question have been provided during the beta-testing phase, but the qualitative responses suggest that the puzzle mechanics still added extraneous cognitive load (Mayer, 2009) for some student learners.

Table 6
Influence of GBL on learning outcomes.

Themes	Sub-themes	Sub sub-themes	Extract	ID
Intrapersonal aspects	Affective	Rewarding	*Being able to solve the puzzles from using the concepts that we learnt*	27
		Engaging, fun	*The game is super super fun! It is a very enjoyable tutorial and it helped to recap a lot of the lecture content covered and apply it.*	144
		Reduced workload	*The best thing was that we did not have to do tutorial as homework beforehand.*	13
		Reduced stress	*the game was a less stressful way of learning concepts we need to grasp.*	49
	Cognitive	Consolidation of learning	*Reinforced the concepts learnt in class, eg boxplot, normal distribution etc.*	109
		Application of concepts	*Interesting way of making us apply the concepts we have learnt in class*	24
		Thinking skills	*Having to put on my thinking caps to solve the mystery case*	50
	Attitudinal	Perspective change	*Learning statistics can be in a more fun manner*	6
		Confronting one's assumptions	*assuming that I knew some of the statistical knowledge and the use of excel*	46
	Behavioural	Technical skills (e.g. excel, data visualisation, data analysis)	*By solving statistical problems, I learnt to apply what I have learnt from the lecture videos. For instance, finding interquartile range using google sheets*	60
		Soft skills (e.g. communication)	*Teamwork is important, we need to share ideas and discuss to better understand certain concepts*	45
Interpersonal aspects	Peer learning	Support from others	*I liked that there was a storyline and that we could work together and help each other out in the game. If I did not know how to do a certain question, my friend would help me out and vice versa.*	30
	Group interaction	–	*Discussion and interaction with teammates*	84
	Camaraderie	Having fun together	*Different team mates have a stronger suit in different sections and we help each other and have fun together.*	119
	Goal orientation	Solving the case/puzzles was rewarding	*The eureka moment when we solved the puzzle; solving the case*	77

From a GBL perspective, however, we argue that extraneous cognitive load attributed to puzzle mechanics cannot always be fully eliminated given the intrinsic nature of the puzzle activity, but considerations to further clarify the puzzle instructions and in-game hints could be made with future implementations of the game.

Although verbal hints from the gamemasters were provided to students as a "helpline", we observed that very few groups used this source of support, and that students tended to prefer trying multiple other ways to solve the puzzles within their groups instead. Similar help-seeking avoidance behaviour has been reported in other studies. Norum et al. (2024) observed that hint requests were used sparingly despite providing hint access to learners in an online mathematics game. Yang et al. (2022) found that students who received instructional videos either before or after a game level in a physics game delayed asking for help, compared to those who did not receive any videos. Even in educational non-game settings, students may not know when to seek help, and may rather attempt to problem-solve independently and delay help-seeking even when resources are available to them (Aleven & Koedinger, 2000). Hence, it is likely that the scaffolds provided (video lectures, puzzle instructions, and in-game hints) in our ER were effective and enhanced learners' sense of competence (Ryan et al., 2006) to attempt independent problem-solving.

4.1. Improvements

While the impact of the ER game on learning was generally positive and affirmed the design of the game, there is room for improvement. Students reported glitches such as a link in the game that allowed a player to skip to a few steps ahead without having to attempt some puzzles, and how clicking on an audio clip caused one player to restart the game. Such technical issues were rare, but were resolved where possible before subsequent runs of the game. Suggestions from participant feedback included integrating all puzzles, clues, and information required for the game within a single platform, and having a synchronised, multi-player "co-op" mode where individual players in a team get different clues and are required to piece the clues or perform a coordinated task together to progress in the game. Although an ideal scenario, these changes would require re-designing major parts of the game or developing the game on a platform that is better able to manage these technical aspects but would require more advanced programming skills (e.g., Unity). To address spatial limitations, bigger classrooms could be reserved (if available) so that groups may be spaced further apart to minimise distractions coming from other groups' discussions during the ER session.

4.2. Practical implications

Findings from this study extend research on the effectiveness of ERs as a pedagogical approach for statistics education, especially for courses with students of diverse disciplinary backgrounds and statistical experience. Aside from knowledge gains and development of communication, teamwork, and social skills, team-based ERs are especially beneficial for classes of students with varied statistical experience as a powerful approach to promote peer learning. Educators are strongly encouraged to consider utilising GBL to improve student perceptions and attitudes towards learning statistics and foster active learning. This study also addresses "return on investment" concerns (Fotaris & Mastoras, 2019) by showing that the development of effective educational games need not be financially costly or require vast resources, since free web-based tools that present low barriers to entry for non-game developers are available, and can be scaled up for large classes, whilst having high reusability value. For educators interested in exploring GBL, based on our

Table 7
User experience in ER.

Themes	Sub-themes	Sub sub-themes	Extract	ID
In-game experience	In-game tasks	Authenticity of assigned tasks for learning	*I learnt how to better apply statistics into a real-world context, or rather better see the uses for it*	95
	Engaging narrative	–	*the storyline was not just any clichéd and predictable storyline but there were twists, turns and baits such that the culprit could only be identified through completing the storyline*	64
Game interface	Interactive game environment	–	*Finding out hidden sections/portions of the story through interacting with the various elements on screen.*	115
	Game design	–	*the whole format and concept of using an escape room to apply what we learnt was really good*	11
	Navigation	Evidence-gathering	*The constant back navigation needed to view the witness statements to correlate with the evidence we have*	5
			having to go back to different pages to look at the clues.	16
		Multiple platforms - messy	*too many pop up items like having to open google forms and then a google drive then excel*	71
Technical issues	Hardware	–	*the computer lab PC took forever to start and launch chrome*	43
	Glitches/bugs	–	*Some of the UI were missing ie. I could see the ending parts without clearing the probability questions that my groupmates faced.*	86

Table 8
Hurdles to learning.

Themes	Sub-themes	Sub sub-themes	Extract	ID
Barriers to problem-solving	Solution roadblock	–	*Not being able to come up with the code when we already had all the puzzle pieces in place.*	103
	Hint system	Hints were not straightforward	*we couldn't figure out some parts even with the hint provided*	102
		Missed out obvious hints	*Sometimes the clues were hidden in plain sight*[3]	21
		Insufficient instructions	*There was little information on which sets of data or clues to use and we only found out when clicking on the hints.*	76
Challenges in team dynamics	Difficulties embracing varied progress	Team members at different speeds	*… some people in the team do not even find the same clue and progress at different speeds*	113
External factors	Timing of activity	–	*The game takes place … when I am quite tired from the day of class.*	9
	Spatial limitations	–	*hearing answers from [other] groups spoil the fun*	52
No hurdles encountered	–	–	*i dont think it was very frustrating, i enjoyed it!*	142

[3] This was a response made to the question "*Please share with us what were the most frustrating part(s) of this educational game.*"

experience in developing this ER, we would like to stress the importance of conceptualising a well-thought-out game design, as well as ensuring proper alignment of the gameplay with learning objectives, for a successful implementation of the GBL experience.

4.3. Strengths, limitations, and future research

Our study addressed frequently cited methodological pain points bemoaned by other researchers over the years. First, the design of the ER was pedagogically grounded and aligned with learning objectives, assessed over three iterations, and with large sample sizes achieved, following which detailed implementation steps and a post-game debrief were described. Importantly, learning gains were objectively measured using pre- and post-tests, and triangulated with student perceptions and gamemaster observations. Additionally, respondent demographics were similar to participant enrolment demographics owing to the high response rates from prompt administration of survey questionnaires. Consequently, results were consistent across cohorts, strengthening the reliability of this study. Moreover, by administering the post-test immediately and one-week post-ER, this allowed us to measure learning acquisition and short-term retention of content knowledge. Conducting the post-test two or three weeks after would have introduced bias since the concepts would have been re-visited in class and any observed effects cannot be attributed solely to the ER. A longer-term evaluation was considered but was not performed, because from prior experience, the response rates are likely to drop significantly with time. Thus, further research assessing longer-term learning gains from ERs could further extend current understanding regarding the potential of ERs for education.

Despite its merits, this study is not without limitations. Meta-analytic evidence suggests that virtual GBL interventions improved learning more than traditional teaching approaches in the analysed studies (Barz et al., 2024; Clark et al., 2016; Wouters et al., 2013). However, as a control group was not included in our study, it was not possible to compare the differences between the ER and traditional instruction on learning. The ER cohorts and the two preceding cohorts (i.e., pre-ER cohorts) did not differ significantly in terms of demographics, grade distribution, and proportion of passes for the quantitative assignment. This was not unexpected given the ER was designed to align with the learning objectives of the *first* lesson of the quantitative section for the quantitative assignment, rather than the overall learning objectives on which students were assessed in the entire quantitative assignment. The pre-ER cohorts received a "traditional" tutorial where students worked on the tutorial questions before coming to class; during the tutorial, the instructor facilitated the discussion by getting students to share their responses before going through the answers. It should be noted that the tutorial for both pre-ER cohorts was conducted remotely over Zoom during the COVID-19 pandemic, while the ER was conducted on-site for all ER cohorts, during the endemic stage. Nevertheless, while the ER was not used as the main teaching approach in the entire quantitative section of the course, our study demonstrates that the ER is effective as a non-traditional pedagogical approach.

Furthermore, in view of the specificity in game design and scope, the generalisability of this study remains limited. The myriad of game designs, methodological approaches used, and educational settings in which ERs have been implemented are hurdles to synthesising strong meta-analytic evidence in this field. The meta-analysis by Lopez-Pernas (2024) reported high heterogeneity despite observing a large positive effect of ERs on learning. While systematic reviews have reported generally positive learning outcomes from educational ERs (Burbage & Pace, 2024; Hintze et al., 2023; Lathwesen & Belova, 2021; Makri et al., 2021; Quek et al., 2024), it was observed that many studies did not use objective quantitative evaluations. Therefore, future studies could consider incorporating objective measures of learning and ideally a control to further strengthen the evidence of actual effectiveness of ERs in statistics education.

4.4. Conclusions

A virtual ER was developed for an introductory statistics course and evaluated among multiple cohorts of undergraduate public health students. Pre-post tests showed significant learning gains with large effect sizes across all cohorts, with positive learning outcomes in the cognitive, affective, behavioural and interpersonal domains evidenced in the qualitative data. While we encountered challenges such as the lack of availability of a larger venue for the activity and in identifying specific glitches reported by students, our findings suggest that ERs can be a powerful approach in fostering peer learning and shifting negative attitudes about difficult subjects such as statistics. Educators keen in exploring ERs in their classrooms are highly encouraged to consider pedagogically-informed game designs and leverage technology in creating engaging and effective learning experiences for their classes. The link to the virtual ER may be provided to interested educators and researchers upon request.

CRediT authorship contribution statement

Woon Chien Cecilia Teng: Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Gek Ling Claire Tan:** Writing – review & editing, Writing – original draft, Project administration, Formal analysis.

Data availability

Data will be made available on request.

5. Appendices

5.1. Figures

Fig. 1. In-game screenshot with details redacted and modified.

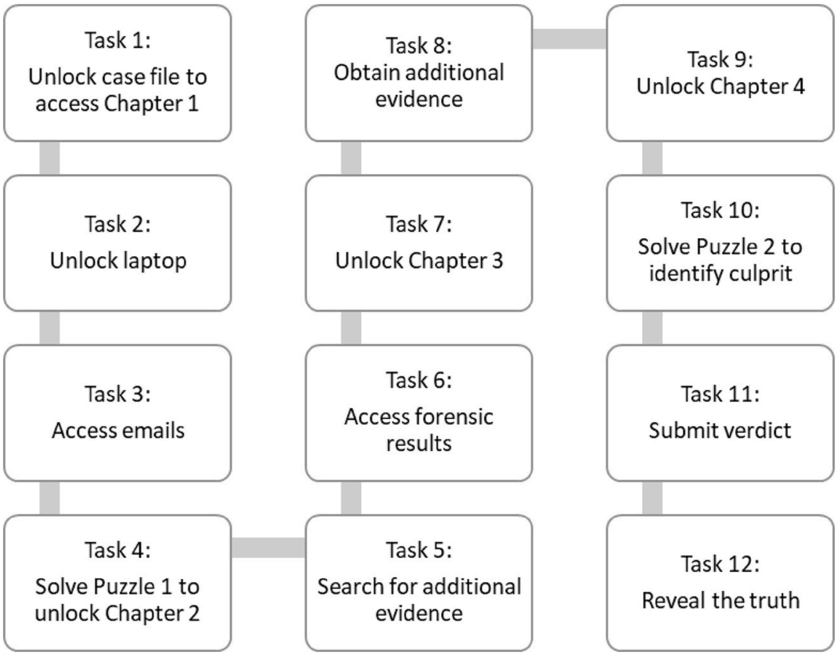

Fig. 2. ER schematic.

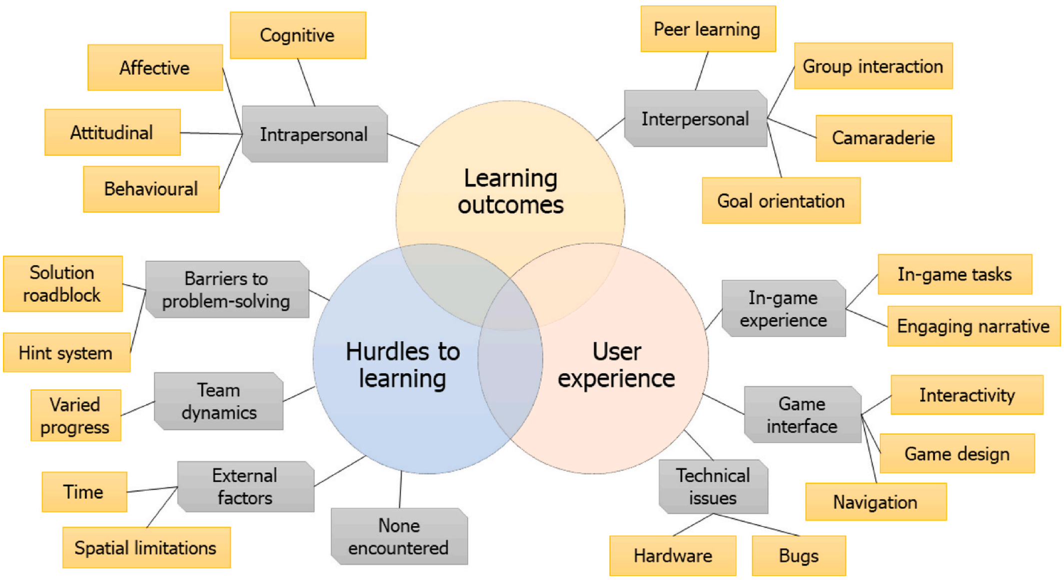

Fig. 3. Thematic map.

5.2. Post-activity Questionnaire

Section A: Demographics

1. Which Faculty/School are you from?
   Arts and Social Sciences
   Business
   Computing
   Design & Environment
   Engineering
   Law
   Music

Science
Others (Please specify:__________)

2. Please indicate your major(s) below.
   Applied Mathematics
   Chemistry
   Communications and New Media
   Economics
   English Language
   Environmental Biology
   Environmental Geography
   Food Science and Technology
   Global Studies
   Life Sciences
   Pharmacy
   Philosophy
   Political Science
   Psychology
   Social Work
   Sociology
   Statistics
   Others (Please specify) _____________
3. Which undergraduate year of study are you currently in?
   1st year
   2nd year
   3rd year
   4th or 5th year
4. I have taken an introductory statistics module before this module.
   Yes
   No
5. I have participated in a game (may not need to have an educational purpose) similar to today's educational game and/or a point-and-click game before today.
   Yes
   No

Section B: Quantitative section

1. What is your overall opinion regarding the difficulty of this educational game?
   Very difficult
   Difficult
   Average
   Easy
   Very easy
2. The learning objectives of this educational game are shown below, and are related to Week 4's content.
   1 Distinguish between nominal, ordinal, ratio, and interval variables.
   2 Distinguish between independent and dependent variables.
   3 Report basic descriptive statistics (e.g. mean, median, quartiles).
   4 Interpret a box plot.
   5 Describe examples of events that follow a binomial distribution.
   6 Determine probabilities using the t-distribution and normal distribution.
   7 Understand the central limit theorem.
   8 Understand how confidence intervals are constructed.
   9 Describe basic sampling methods.
3. To what extent do you agree or disagree with the following statements?
   (1 = Strongly Disagree to 5 = Strongly Agree)

Statements	1	2	3	4	5
1. The educational game aligns with the learning objectives above.					
2. I feel that I would learn more from this educational game than if the same learning objectives were to be covered in a traditional tutorial.					
3. The educational game was an effective way to reinforce the related topics covered in Week 4's on-site and pre-recorded lectures.					
4. I enjoyed the educational game.					
5. The educational game was a stressful experience.					
6. I enjoyed working with my team in the educational game.					
7. The educational game was more engaging than a traditional tutorial.					
8. The main difficulty of the educational game lied in mastering the course materials.					
9. The main difficulty of the game lied in the non-educational game mechanics.					
10. The educational game was well-designed.					
11. The detective-themed storyline was engaging.					
12. The time given for the educational game was adequate.					
13. I benefited from the teamwork in my group.					
14. I learnt about statistics and/or quantitative research methods from my team member(s).					
15. I would rather prefer participating in the educational game alone than in a team.					
16. All the members of my team were equally involved in solving the different puzzles.					
17. My team mates helped each other out in the educational game.					
18. I would rather have been part of a larger team.					
19. The post-activity reflection and debrief were helpful.					
20. The hints provided within the game platform were useful to progress in the game.[a]					

[a] Respondents were allowed to indicate "NA" if they did not refer to any hints during the gameplay.

Section C: Qualitative section

Learning takeaways.

1. Please share with us what you have learnt from participating in this educational game.

User experience feedback.

2. Please share with us what were the most enjoyable part(s) of this educational game.

3. Please share with us what were the most frustrating part(s) of this educational game.

4. Please share with us how the educational game could be improved for future students.

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